C1 Four Topology MOSbius

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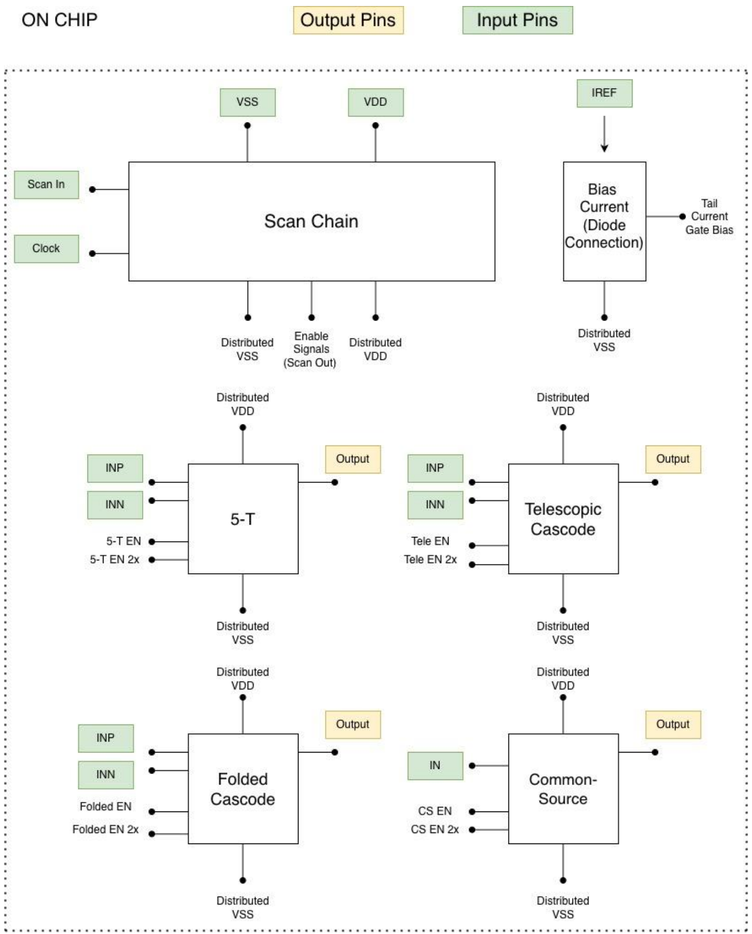
Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
boibun.	Columbia University	Undergraduate	Team Member
maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member

Project Information

Goal: To design and fabricate a chip containing several of the most common amplifier topologies, all with accessible pins, so that a student can wire up, test, characterize, and compare each amplifier. Write a short lab manual to accompany the chip (consider during design, but can follow design/layout submission). Students measure DC gain, GBW, slew rate, stability, PSRR, etc. for each amplifier, learning the trade-offs through hands-on experimentation

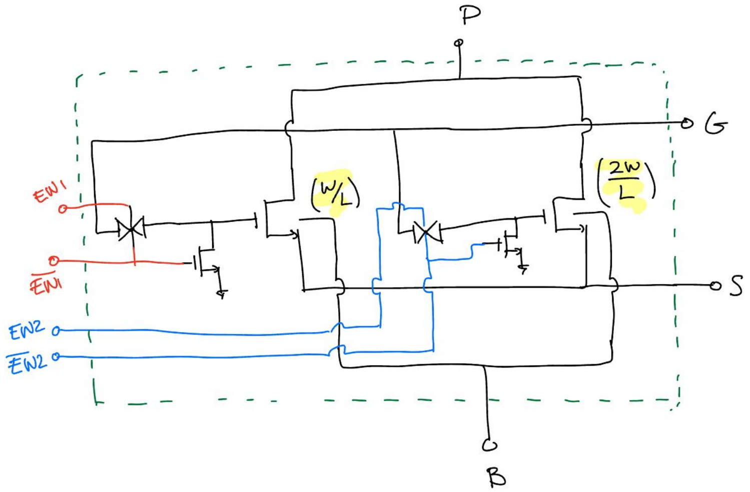
Block Diagram

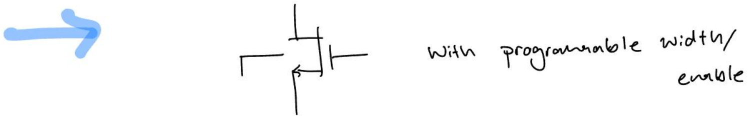
- All amplifiers are NMOS input
- IREF comes from an off-chip current reference and is driven through a diode-connected NMOS to generate a gate bias voltage. This voltage is distributed to the tail current sources of all amplifiers, enabling their operation
- Scan In is an off-chip input that encodes which enable signals we want active, and Clock is an off-chip clock input. Both are generated by a Raspberry Pi–style device, using an implementation similar to the original MOSbius



Programmable Transistor Implementation

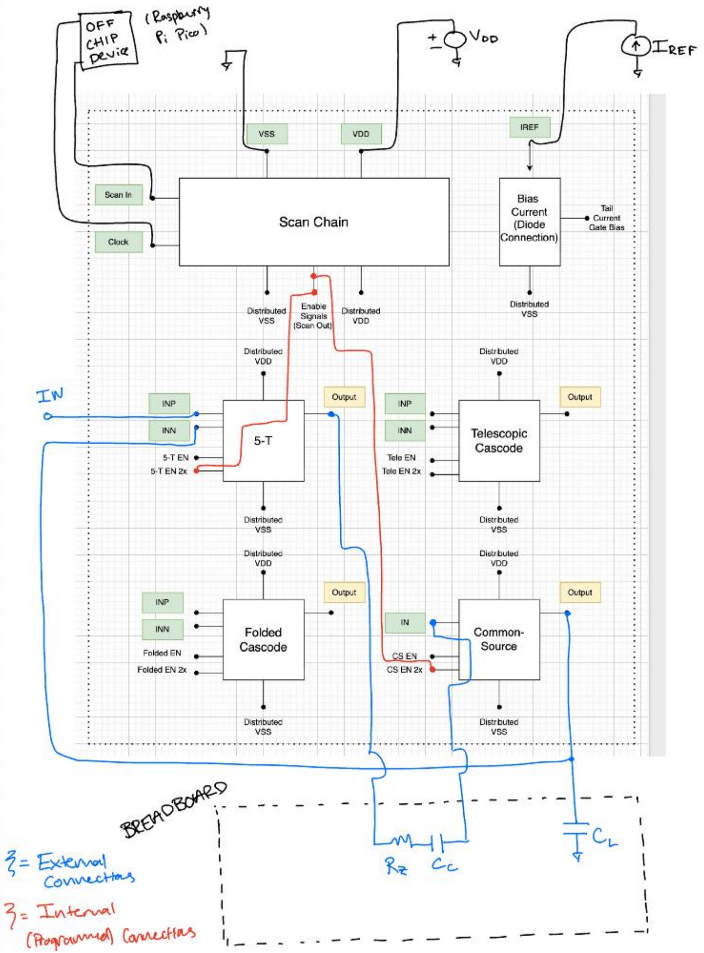
- The enable signals turn each amplifier on by connecting the gate bias voltage to its tail current transistor (via transmission gates shown). With no enable active, the bias can't reach the tail (so the amplifier is off)
- We use a programmable transistor similar to the one shown at right. The complementary EN1 signal selects default sizing, EN2 selects 2× sizing, and enabling both EN1 and EN2 gives 3× sizing





Example Configuration

- Unity Gain Buffer, 5-T first stage, CS second stage, with 2x sizing
- Distributed VDD and VSS connections are not shown; these pins are always tied to the corresponding pin of each amplifier, even when it is not enabled
- The Tail Current Gate Bias connection is also not shown; asserting the 2× enable internally connects this pin to the gate of the 2×-sized tail current transistor



Pin List

Total Pins: 16

Digital Input (2): Scan In, Clock

Analog Input (10): VDD, VSS, IREF, IN(CS), INP/INN (5T), INP/INN (Telescopic), INP/INN (Folded)

Analog Output (4): OUT (CS), OUT (5T), OUT (Telescopic), OUT (Folded)

Control (Transistor Enable) Bits

Each transistor will have two bits to choose size (00 - 0x, 01 - 1x, 10 - 2x, 11 - 3x). Each amplifier will receive these two bits from the scan chain. Therefore, the scan chain will be 8 bits long and hold the bits in order when the clock is shut off. Listed below is the order of the bits as they come off the scan chain. 0 - being first flip flop, 7 - last flip flop

Enable[0-1] = 5T transistors
Enable[2-3] = Telescopic Cascode transistors
Enable[4-5] = Folded Cascode transistors
Enable[6-7] = Second Stage transistors

Amplifier Specs

- Each amplifier targets 1 MHz GBW into a 100 pF off-chip load at default sizing
- The user is free to choose any off-chip load capacitance; since GBW depends directly on load capacitance, it is partially user-programmable
- Sizing also affects GBW: at the fixed overdrive set by the current mirror, tripling W/L triples both the bias current and gm (gm/id point constant), raising GBW proportionally
- No on-chip capacitors. All compensation and load capacitors are implemented off-chip on a breadboard, as shown in the example configuration

Initial Ideas for Amplifier Experiments

- Unity-gain buffers (dynamic behavior/range)
- Inverting amplifiers (static accuracy and gain/gain error)
- LDO / voltage regulators (load regulation and supply rejection)
- Active filters (GBW)

References

https://github.com/AndrewwChon/PLL_MOSbius

https://github.com/AutoMOS-project/AutoMOS-chipathon2025

https://mosbius.org/0_front_matter/intro.html